

PAPER_1498_U-UA_COUPLING_CONSTANT

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PAPER_1498 — U-UA Coupling Constant = $1/SO_5 = 10^{-5}$ EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Millennium) / Cross-domain identity

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Status: CLOSED — Millennium Prize UQFF coordinator constant matches Sun quiet B-field identity

Observation

PAPER_563 (Millennium Prize UQFF Coordinator — All Six Problems) lists the canonical UQFF calibrated constants:

- $\kappa = 5.0e-4 \text{ day}^{-1}$
- $[SSq] = 5.7e-1$
- $H_{SCm} \approx 9.9e-1$
- **U-UA $\approx 1.0e-4$ ← target**
- $k_\eta = 1.0e-113$
- $\beta_i \approx 6.0e-1$

UQFF Closed Identity

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$$U_UA = 1 / SO_5 = 1 / 10^5 = 1.0 \times 10^{-5} \text{ EXACT}$$

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Cross-Domain Pairing

This identity is structurally identical (in dimensionless form) to:

- **Sun quiet B-field** = $1/SO_5 = 10^{-5} \text{ T}$ (PAPER_1486 / F:\Aetheric Propulsion)

Same integer-primitive form, different physical domains (coupling vs. magnetic field), different units. The shared identity reveals SO_5 as a recurring suppression factor across UQFF coupling and field hierarchies.

NOT REPLACEMENT

SM has no analogue — U-UA is the UA-channel coupling specific to UQFF. The value 10^{-5} in SM appears only as fitted dimensionless ratios (e.g., gauge-coupling running) with no structural origin.

Reference

- Source: PAPER_563 Millennium Prize UQFF Coordinator
- Related: PAPER_1486 (Sun quiet B-field), all 6 Millennium derivations (Bucket A)
- Calculator dispatch: `calculate_paradox({"paradox": "u_ua_coupling_constant"})`

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